

Proxy Hardening — Test-Type Coverage Status

Revision: 3 **Last modified:** 2026-07-01T13:26:00Z **Status:** In progress — 9 test types PASS with captured live evidence (unit [control-plane, all pkgs pass], stress+chaos, DDoS, concurrency/atomicity, memory-soak, **P10 VPN fail-closed full-automation GREEN, race/deadlock [0 DATA RACE], benchmarking/performance [data-plane p50=86ms + control-plane Go micro-benchmarks], Challenges [2 PASS / 1 honest SKIP]**), 2 honest SKIP (security ACL — no autonomous real-ACL-deny topology; HelixQA — bank blocked on 6 un-vendored siblings), integration + e2e PENDING (broader tests/run-tests.sh suite). The control-plane store-layer audit-atomicity TOCTOU was investigated this round and found already CLOSED (store.WithTx), verified RED→GREEN. No PASS is claimed without a real qa-results/ evidence path (§11.4.6). **Authority:** Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc reconciling the §11.4.169 test-type-coverage matrix for the proxy hardening workstream. **Companion:** summary [Status_Summary.md](#)

Operator-blocked / pending items (read first — §11.4.45 O(1) surface)

Item	Why blocked / pending	Unblock condition
P10 real-VPN egress-half (SKIP)	The egress half — proving traffic actually EXITS via the tunnel — needs operator gluetun WireGuard credentials (§11.4.21). The fail-closed half is now GREEN.	Operator provides gluetun WireGuard creds; then the real-egress positive path can be captured.
security ACL deny-does-not-leak (SKIP)	No autonomous topology on the base stack produces a real Squid ACL deny (observed deny code 000000 — cannot prove deny nor leak).	A dynamic/ACL-configured topology where a CONNECT to a denied target returns 403/407, so deny-without-leak is provable.

§11.4.169 test-type-coverage matrix (captured-evidence-driven — §11.4.5/ §11.4.69)

Base proxy stack UP on :53128 at capture time. Every PASS below cites a real, non-empty qa-results/ artefact captured live this session.

Test type	Status	Evidence path	Notes
Unit	PASS	qa-results/unit/control-plane_unit_20260701T131306Z.log	go tes plane: 100%, 81.2% 64.5% only).
Integration	PENDING	—	Not re harder
E2E	PENDING	—	Not re harder
Full-automation (P10 VPN fail-closed)	PASS	qa-results/dynamic/vpn_failclosed/20260701T130115Z/verdict.txt (+ req_1..3.body branded pages)	tests/ booted ERR_TU marke PID un polarit Egress
Challenges	PASS (2/3, 1 honest SKIP)	qa-results/challenges/20260701T131440Z/summary.txt	challe proxy: direct cache Age ca §11.4.
HelixQA	SKIP (blocked)	qa-results/helixqa/20260701T090532Z/(honest SKIP path)	Hones — subn not ve llm_pr Proxy cases 6 sibli
DDoS / load-flood	PASS	qa-results/ddos/proxy_flood_20260701T122320Z/flood.evidence (+ census.txt)	300/30 timeou listeni
Security	SKIP	qa-results/security/proxy_acl_20260701T120913Z/s1_acl_deny.evidence	Hones real-A deny o (s2_he (§11.4

Test type	Status	Evidence path	Notes
Stress + Chaos	PASS	qa-results/stress/ proxy_forward_20260701T120843Z/ stress.evidence · qa-results/chaos/ proxy_restart_20260701T120941Z/ recovery_trace.log	Stress concu proxy- recove
Concurrency / atomicity	PASS	qa-results/concurrency/ proxy_concurrency_20260701T122740Z/ concurrency.evidence (+ clients.tsv) · qa- results/unit/control- plane_toctou_atomicity_20260701T131522Z.log	Data-p socks5 OVERAL layer 7 fails) i mutat Begin TestAt polariz
Race / deadlock	PASS	qa-results/race/control- plane_race_20260701T125739Z.log	go tes incl. c compi {aclhe EXIT= spawn (inter tunnel layer a round atomic
Memory	PASS	qa-results/memory/ proxy_soak_20260701T122643Z/soak.evidence (+ rss_timeseries.tsv)	RSS-b 000 B
Benchmarking / performance	PASS	qa-results/benchmark/ proxy_forward_20260701T130414Z/latency.txt · qa-results/benchmark/control- plane_gobench_20260701T131745Z.log	Data-p live vs p95=0. OVERAL now c (0 allo Compil µs/op.

Honest boundary (§11.4.6)

This matrix reconciles the **hardening** test-type picture proven in this session. Nine types (unit, stress+chaos, DDoS, concurrency/atomicity, memory, P10 fail-closed full-automation, race/deadlock, benchmarking, Challenges) are PASS with captured live qa-results/ evidence. Two SKIPS are honest (§11.4.3): security ACL (no autonomous deny topology on the base stack) and HelixQA (bank blocked on 6 un-vendored own-org siblings — the runner + proxy.yaml bank exist; unblock = vendor them or pass HELIXQA_BIN).

Integration + e2e remain PENDING — not claims the capability is broken; covered by the broader tests/run-tests.sh suite. The two follow-ups flagged earlier this round are RESOLVED: the store-layer audit-atomicity TOCTOU was found already CLOSED (store.WithTx, verified RED→GREEN), and the control-plane Go-benchmark gap is closed (3 new *_bench_test.go). None of this substitutes for the §11.4.40 full-suite retest before a release tag.